

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

MLT

Section Id :	640653138922
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653387842
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532128321 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING

TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536934829. ✓ Yes

6406536934830. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653387843

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532128322 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a dataset with two data-points for a binary classification problem with features in $\mathbb{R}^2$. $(1, 0)$ is a data-point from the positive class while $(-1, 0)$ is from the negative class. We now consider three linear classifiers: perceptron, logistic regression and SVM.

Use gradient ascent to train the logistic regression model with $\eta = 1$. For both perceptron and logistic regression, run exactly one iteration of their respective algorithms starting with $w^{(0)} = (0, 0)$ in each case. That is, you should stop after you get $w^{(1)}$. Train the SVM in the usual way.

Use 0 as the negative class label for logistic regression and -1 for SVM and perceptron. Use the following rule for determining the predicted labels for the three classifiers:

$$\begin{array}{c|c} \text{SVM and Perceptron} & \text{Logistic Regression} \\ \hline \hat{y} = \begin{cases} 1, & w^T x \geq 0 \\ -1, & w^T x < 0 \end{cases} & \hat{y} = \begin{cases} 1, & w^T x \geq 0 \\ 0, & w^T x < 0 \end{cases} \end{array}$$

If $w^* \in \mathbb{R}^2$ is the weight vector with the largest margin, which of the following is true?

Options :

6406536934831. ✓ All three classifiers output w^*

6406536934832. ✗ Only SVM outputs w^*

6406536934833. ✗ Only SVM and perceptron output w^*

6406536934834. ✖ Only perceptron and logistic regression output w^*

Sub-Section Number : 3
Sub-Section Id : 640653387844
Question Shuffling Allowed : Yes

Question Number : 3 Question Id : 6406532128323 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset in $\mathbb{R}^2$ defined as

$$D = ([1, 1]^T, +1), ([2, 1]^T, +1), ([1, -1]^T, -1), ([2, -1]^T, -1)$$

Which of the following weight vectors $w \in \mathbb{R}^2$ can correctly separate the two classes using a linear decision boundary? (Select all that apply.)

Options :

6406536934835. ✔ $w = [0, 1]^T$

6406536934836. ✖ $w = [1, 0]^T$

6406536934837. ✔ $w = [1, 5]^T$

6406536934838. ✖ $w = [-1, 1]^T$

Question Number : 4 Question Id : 6406532128330 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of n datapoints $x_1, x_2, \dots, x_n$, where each $x_i \in \mathbb{R}^4$. Let the four features be denoted by f_1, f_2, f_3 , and f_4 .

Suppose the features satisfy the following linear relationship:

$$f_3 = 2f_1 + 3f_4$$

That is, one of the features is an exact linear combination of others. A linear regression model is trained to predict a target variable

y using these features. Consider the following statements:

Which of the following is (are) true.

Options :

6406536934850. ✓ Linear regression without regularization will achieve training MSE less than or equal to that of ridge regression.

6406536934851. ✓ The solution for linear regression without regularization may not be unique.

6406536934852. ✗ Ridge regression (L2 regularization) will always set at least one coefficient exactly to zero.

6406536934853. ✓ Lasso regression (L1 regularization) can force some feature weights to become exactly zero.

Sub-Section Number :

4

Sub-Section Id :

640653387845

Question Shuffling Allowed :

Yes

Question Number : 5 Question Id : 6406532128324 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a binary classification dataset in $\mathbb{R}^4$ that is linearly separable. Suppose there exist two distinct weight vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ that both correctly classify all training points with a positive margin.

The Perceptron algorithm is applied to this dataset.

Which of the following statements are correct? (Select all that apply.)

Options :

6406536934839. ✗ The Perceptron algorithm is guaranteed to converge to $\mathbf{w}_1$.

6406536934840. ✓ The final weight vector obtained by the Perceptron algorithm depends on the initialization and the order in which the data points are presented.

6406536934841. ✓ The Perceptron algorithm is guaranteed to converge to some separating hyperplane in a finite number of updates.

6406536934842. ✗ The Perceptron algorithm always converges to the maximum margin separator among all possible separating hyperplanes.

Sub-Section Number :

5

Sub-Section Id :

640653387846

Question Shuffling Allowed :

Yes

Question Number : 6 Question Id : 6406532128325 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 4

Question Label : Short Answer Question

Consider a hard-margin SVM trained on a linearly separable dataset. The optimal Lagrange multipliers are

$$\alpha = [2 \quad 0 \quad 3 \quad 0]^T.$$

The training dataset $(\{(\mathbf{x}_i, y_i)\}_{i=1}^4 \subset \mathbb{R}^2)$ is given by

$$D = \{((1, 2), +1), ((2, 2), +1), ((2, -1), -1), ((1, -2), -1)\}.$$

Given the test point $\mathbf{x}_{\text{test}} = (1, 1)$, determine the predicted label $\hat{y} \in \{+1, -1\}$. Enter the answer as either 1 or -1

Hint: The prediction for a test point $\mathbf{x}_{\text{test}}$ is given by

$$\hat{y} = \text{sign}((\mathbf{w}^*)^T \mathbf{x}_{\text{test}})$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 7 Question Id : 6406532128331 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 4

Question Label : Short Answer Question

Consider a binary classification problem using logistic regression. The model predicts the probability of class $+1$ as:

$$P(y = +1 \mid x) = \frac{1}{1 + e^{-w^T x}}$$

A threshold of 0.7 is used for classification. The predicted class label is given by:

$$P(y \mid x) = \begin{cases} +1, & \text{if } P(y = +1 \mid x) \geq 0.7 \\ -1, & \text{otherwise} \end{cases}$$

The values of $w^T x$ for the following datapoints are:

$x_1 : -1.2 \quad x_2 : 0.0 \quad x_3 : 2.5 \quad x_4 : 0.4 \quad x_5 : 0.8$

How many datapoints will be classified as class $+1$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 8 Question Id : 6406532128338 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 4

Question Label : Short Answer Question

A coin is flipped 100 times, and it lands on heads 60 times. You use a Beta(10, 40) prior to estimate the true probability of heads. Given these observations, calculate the posterior mean of the probability of heads. Enter the answer correct to three decimal places.

Note:

$$f(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq \theta \leq 1, \alpha, \beta \geq 0, \quad \text{where } B(\alpha, \beta)$$

• Pdf of Beta distribution is
is the Beta function.

• The mean of the Beta distribution is: $E[\theta] = \frac{\alpha}{\alpha+\beta}$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.44 to 0.49

Sub-Section Number :

6

Sub-Section Id :

640653387847

Question Shuffling Allowed :

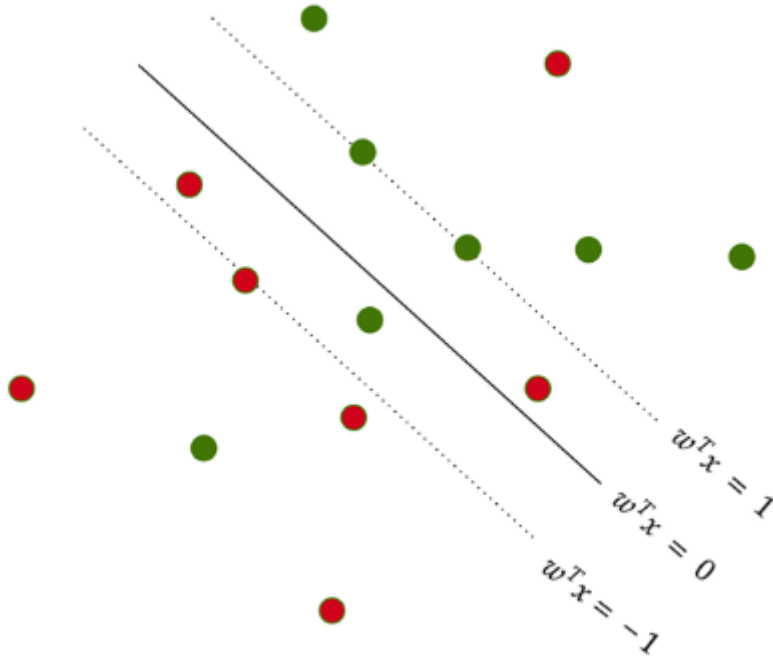
No

Question Id : 6406532128326 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (9 to 10)

Question Label : Comprehension

Consider the following soft-margin Support Vector Machine (SVM) classifier. The solid line represents the decision boundary $w^T x = 0$, and the dotted lines represent the margin boundaries $w^T x = 1$ and $w^T x = -1$.

Red points correspond to the negative class, and green points correspond to the positive class.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 9 Question Id : 6406532128327 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many datapoints will incur a penalty (i.e., have non-zero slack variables $\xi_i > 0$)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 10 Question Id : 6406532128328 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many datapoints will definitely have $\alpha_i = 0$ in the dual formulation of SVM?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Id : 6406532128332 Question Type : COMPREHENSION Sub Question Shuffling

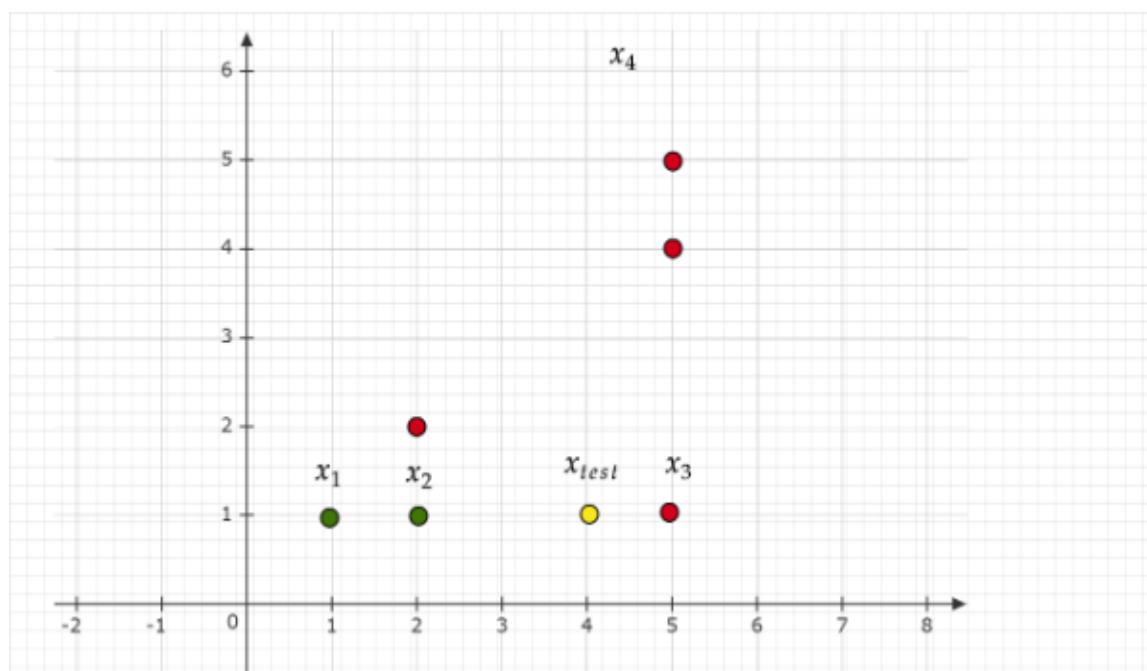
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (11 to 12)

Question Label : Comprehension

Consider the given 2D diagram. The datapoints are slightly exaggerated for better visualization.

Red points represent the negative class (-1) and green points represent the positive class ($+1$). A test point x_{test} is shown. Use the K-Nearest Neighbors (KNN) algorithm with $K = 3$.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 11 Question Id : 6406532128333 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Using Euclidean distance, what is the predicted class of x_{test} ?

Options :

6406536934855. ✖ +1

6406536934856. ✔ -1

6406536934857. ✖ Tie

6406536934858. ✖ Cannot be determined

Question Number : 12 Question Id : 6406532128334 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following best explains the classification of x_{test} ?

Options :

6406536934859. ✖ The nearest neighbors are dominated by positive class points.

6406536934860. ✔ The nearest neighbors are dominated by negative class points.

6406536934861. ✖ Positive class points are closer in all directions.

6406536934862. ✔ The classification is determined by majority voting among the nearest neighbors.

Sub-Section Number :

7

Sub-Section Id :

640653387848

Question Shuffling Allowed :

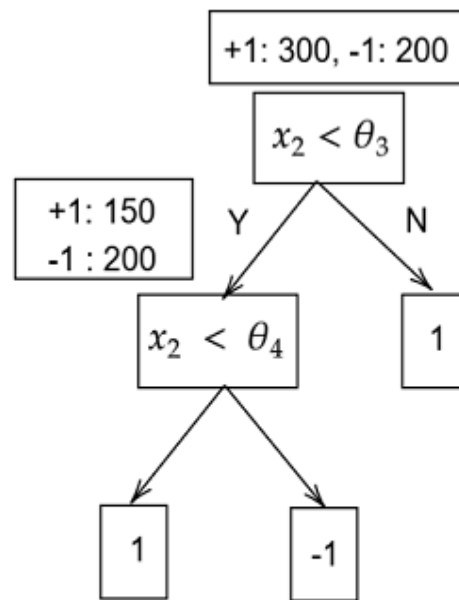
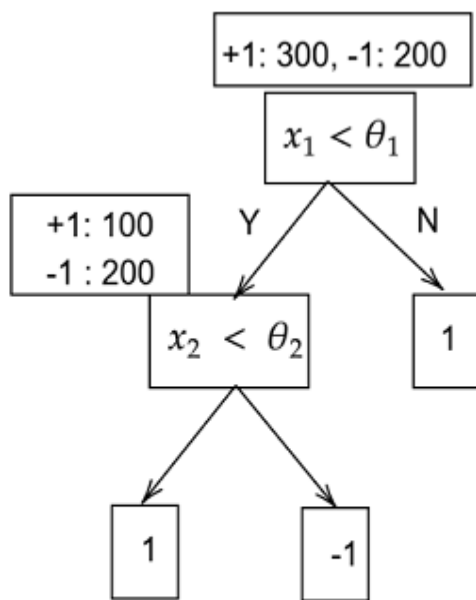
Yes

Question Number : 13 Question Id : 6406532128329 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the two decision trees shown in the figure.



Which of the following statements is correct?

Options :

- 6406536934846. ✓ Tree 1 has higher information gain than Tree 2
- 6406536934847. ✗ Tree 2 has higher information gain than Tree 1
- 6406536934848. ✗ Both trees have equal information gain
- 6406536934849. ✗ Cannot be determined without further splits

Question Number : 14 Question Id : 6406532128335 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following statements is correct?

Options :

- 6406536934863. ✓ Bagging reduces variance, Boosting reduces bias
- 6406536934864. ✗ Bagging reduces bias, Boosting reduces variance
- 6406536934865. ✗ Both reduce only variance
- 6406536934866. ✗ Both reduce only bias

Question Number : 15 Question Id : 6406532128336 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the datapoints $(-2, -2)$ with label $+1$ and $(2, 2)$ with label -1 . A hard margin SVM is applied on this dataset. Which of the following is the decision boundary for the above dataset?

Options :

- 6406536934867. ✗ $x + 2y = 0$

6406536934868. ✓ $-x - y = 0$

6406536934869. ✗ $x - y = 0$

6406536934870. ✗ $y - x = 0$

Question Number : 16 Question Id : 6406532128337 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a dataset $x_1, x_2, \dots, x_n$, where each $x_i \in \mathbb{R}^5$. Principal Component Analysis (PCA) is applied to this dataset and all the five principal components are retained. Which of the following statements is correct?

Options :

6406536934871. ✗ PCA has reduced the dimensionality of the dataset because the combinations of features are de-correlated

6406536934872. ✓ PCA has not reduced the dimensionality of the dataset because all principal components are retained.

6406536934873. ✗ PCA always reduces the dimensionality of the dataset regardless of the number of components retained.

6406536934874. ✗ PCA can never be used for dimensionality reduction.

Sub-Section Number :

8

Sub-Section Id :

640653387849

Question Shuffling Allowed :

Yes

Question Number : 17 Question Id : 6406532128339 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Consider a regression problem in which we use a neural network with the following architecture:

- Input layer has ⁵ neurons
- Three hidden layers, each with ¹⁰ neurons
- Output layer with one neuron

Find the total number of parameters in the network. Ignore the biases.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

260